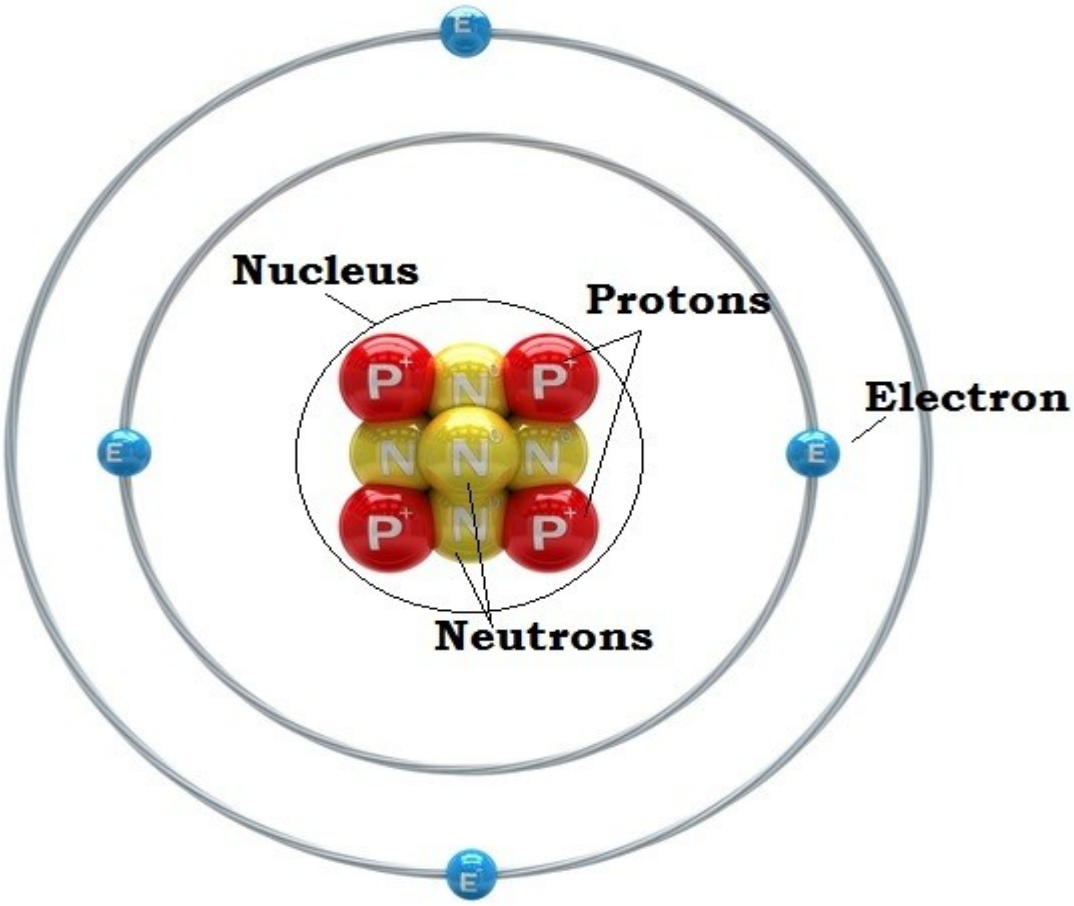
IONS

Atoms contain 3 types of particles: Protons, neutrons and electrons.



Sometimes electrons can be lost or gained by an atom.

When atoms loose or gain electrons they become charged.

There are two possible charges: positive or negative.

Positive = +

Negative = -

When atoms become charged we call them IONS.

Ions are charged particles.

<u>When atoms lose electrons they become positive ions.</u>

Metal atoms tend to lose electrons and become positive ions.

Element	Symbol for element	Symbol for ion	Number of Electrons lost by each atom
Potassium	K	K	1
Lithium	Li	Li	1
Silver	Ag	Ag	1
Calcium	Ca	Ca	2
Magnesium	Mg	Mg	2
Aluminium	Al	Al	3
Copper	Cu	Cu	
Sodium	Na	Na	
Iron (III)	Fe		3
Zinc			2

When atoms gain electrons they become negative ions.

Non-metal atoms tend to gain electrons and become negative ions.

Also when non-metals become ions the end of their name is changed to ide.

Element name	Ion name
Bromine	Bromide
Nitrogen	nitride

Element Name	Ion name	Symbol for element	Symbol for ion	Number of Electrons gained by each atom
Sulphur	Sulphide	S	S	2
Chlorine	Chloride	Cl	Cl	1
Iodine				1
Oxygen		O	O	2